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Synergist Effects on Stability and Friction in N_2/CO_2 Hybrid Foam as a Fracturing Fluid

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Abstract

CO_2 foam fracturing is a promising technology for developing unconventional reservoirs, but its application in deep, high-pressure/high-temperature (HPHT) wells is severely hindered by the inherent instability of CO_2 foam. Conversely, while highly stable, N_2 foam generates excessive friction pressure. This study systematically investigates the potential of N_2/CO_2 mixed gas foams as a high-performance, tunable fracturing fluid. Through a dual-scale experimental approach under simulated reservoir conditions (100°C, 30 MPa), we combined macroscopic performance tests (stability, rheology, pipe friction) with micro-scale mechanistic analysis (interfacial rheology, microscopic morphology). The results reveal a profound synergistic effect. The addition of N_2 does not significantly alter the static interfacial tension but dramatically enhances the interfacial dilational viscoelasticity. Specifically, adding 50% N_2 increased the interfacial dilational viscoelasticity by nearly 6-fold compared to pure CO_2 foam. This enhanced Marangoni effect, coupled with the "trapped gas effect" that suppresses Ostwald ripening, fundamentally improves foam stability. Macroscopically, this was evidenced by the half-life increasing from 20 min (pure CO_2) to over 500 min with just 35% N_2. This enhanced stability directly translates to superior rheological properties; the effective viscosity (at 10 s^{-1}) of a 50% N_2 mixed foam reached 160 mPa·s, a 3-fold increase over pure CO_2 foam (55 mPa·s). However, this performance gain comes at the cost of increased friction, with the friction factor rising from 0.03 to 0.22 (at 0.6 m/s) for the same foams. Based on this comprehensive dataset, we conclude that N_2/CO_2 mixed foam is a highly tunable system. An optimal engineering window exists where N_2 content is between 35% and 50%, which provides sufficient stability and viscosity for effective fracturing while keeping friction pressures manageable. This work provides a quantitative design basis for optimizing N_2/CO_2 mixed foam fracturing fluids, balancing performance and cost to advance the technology's field-scale application.

Keywords: N_2/CO_2 Mixed Foam, Foam Fracturing, Foam Stability, Interfacial Rheology, Synergistic Effect

Introduction

Driven by the dual imperatives of global energy demand and carbon neutrality goals, Carbon Capture, Utilization, and Storage (CCUS) and Enhanced Oil Recovery (EOR) have become pivotal strategies in the energy sector. Gas flooding, a core technology for both, utilizes gases like CO_2 or N_2 to improve oil recovery and achieve geological sequestration. However, this technique faces a fundamental challenge: the significant density and viscosity contrasts between the injected gas and reservoir fluids (oil, water) lead to severe gravity override and viscous fingering. These phenomena drastically reduce sweep efficiency, leaving substantial oil resources unrecovered and geological storage capacities underutilized (Abdelaal et al., 2020, 2021); To overcome this bottleneck, injecting gas in the form of foam is widely recognized as the most effective mobility control technique(Du et al., 2019; Abdelaal et al., 2023; Koyanbayev et al., 2024). Concurrently, foam fracturing has emerged as a superior alternative to conventional water-based fracturing for unconventional reservoirs, addressing challenges like high water consumption and formation damage while improving proppant transport(Wanniarachchi et al., 2017; Yekeen et al., 2018; Yu and Kanj, 2022). CO_2 is highly favored for its miscibility with oil, viscosity reduction, and sequestration value. However, under HPHT reservoir conditions, supercritical CO_2 (SC-CO_2) forms inherently unstable foams. Studies confirm this, showing CO_2 foam is significantly weaker than N_2 foam, with a mobility reduction factor (MRF) of only 3-6 compared to 50-75 for N_2 (Lv et al., 2016; Roozbahani et al., 2024) . This instability is driven by CO_2's high solubility and diffusivity, which accelerates gas diffusion (Ostwald ripening) and leads to structural collapse (Façanha et al., 2022; Gao et al., 2025). Overcoming this instability is a critical bottleneck for EOR and CCUS applications. Conversely, while highly stable, N_2 foam's high viscosity creates excessive injection friction, posing its own operational challenges(Aarra et al., 2014; Gajbhiye, 2021).

To address this instability, academia and industry have explored various technical approaches, primarily categorized as "modifying the liquid phase" and "strengthening the interface". The most prevalent approach involves adding functional agents to the aqueous phase to enhance the strength and resilience of the liquid lamellae. These include polymers for viscosity enhancement (Polymer-Enhanced Foam, PEF), viscoelastic surfactants (VES) that form worm-like micellar networks, and novel, environmentally friendly additives like regenerated cellulose (RC) or glycerol (Eide et al., 2020; Alooghareh et al., 2022; Lv et al., 2025; Zheng et al., 2025). Another strategy focuses on strengthening the gas-liquid interface itself, for instance, by using nanoparticles (NPs) to form a robust "armor" via the Pickering effect (Lv et al., 2025) or by optimizing binary surfactant systems to improve interfacial properties under harsh conditions (Nowrouzi et al., 2025). While these methods have shown some success, they share a common trait: they are "indirect" solutions that attempt to constrain the unstable CO_2 gas phase by reinforcing the liquid or interfacial components. Such approaches often face challenges including high costs, complex formulations, sensitivity to reservoir conditions, and the risk of formation damage.

In contrast to these indirect control strategies, a more direct and fundamental approach is to modify the gas phase itself. Given the excellent stability of N_2 foam under high pressure, mixing N_2 as a stable "skeletal gas" with CO_2 to co-construct the foam system offers a promising solution to fundamentally address the instability of CO_2 foam(Amani et al., 2025). This concept has received initial validation. The pioneering work of (Siddiqui and Gajbhiye, 2017) first demonstrated at the oil-free core scale that adding a small amount of N_2 (5-20%) significantly increases the flow resistance of CO_2 foam. Subsequent studies by (Abdelaal et al., 2020) and (Façanha et al., 2022) applied this technique to oil-bearing core floods, not only confirming the enhanced foam strength but also revealing its significant potential for improving oil recovery. They highlighted that operational parameters such as foam quality, injection rate, and the N_2/CO_2 ratio are critical for optimization.

Synthesizing the existing literature, it is evident that while the N_2/CO_2 mixed foam technology shows great promise, critical knowledge gaps remain: Lack of Multi-Scale Mechanistic Insight, Absence of

Universal Design Criteria, and Low Technology Readiness Level (TRL). Therefore, this study aims to address these gaps. We employ a dual-scale experimental approach, combining macroscopic and microscopic analyses, to systematically evaluate the performance of N_2/CO_2 mixed foams under simulated deep reservoir conditions (100°C, 30 MPa). The core objectives of this research are: (1) To quantitatively reveal the synergistic stabilization mechanism of N_2 on CO_2 foam from the perspectives of interfacial rheology and microscopic morphology; (2) To establish a comprehensive database linking gas composition (N_2/CO_2 ratio), foam quality, and flow conditions (shear rate/flow velocity) to macroscopic properties (stability, viscosity, and friction factor); and (3) To propose a quantitative engineering design criterion for optimizing the N_2/CO_2 ratio, balancing performance requirements (viscosity) with operational constraints (friction), thereby advancing this promising technology towards practical application.

Materials and methods

Experimental materials and devices

Experimental materials: Fumed silica nanoparticles (SiO_2, purity > 99.8%) were sourced from Wacker Chemie AG (Germany). These nanoparticles featured an average particle diameter of approximately 12 nm, a specific surface area of approximately 120 m²/g, and a particle density of approximately 2200 g/L. The particle surface was hydrophobically modified with polydimethylsiloxane (PDMS), resulting in a surface silanol group density of less than 0.5 groups/nm². The foaming agent, sodium dodecyl sulfate (SDS), was provided by Macklin. Industrial-grade CO_2 (purity ≥ 99.5%) and nitrogen (N_2, purity ≥ 99.9%), supplied by Huanyu Jinghui Gas Technology Co., Ltd., were used as the input gases for foam generation. Ethanol (purity > 99.7%), from Shanghai Titan Scientific Co., Ltd., was used as a co-solvent for the nanoparticles. Deionized (DI) water was used throughout the experiments.

Devices: The experimental setup comprised a high-temperature, high-pressure (HTHP) Waring Blender foam generator, a foam viscosity and friction measurement apparatus, a Teclis Tracker-H interfacial rheometer, a Keyence VHX-5000 digital microscope, a METTLER TOLEDO electronic balance, a JULABO water bath, a DJ1C stirrer, and a 2000 W Hangzhou Chenggong YPS17 ultrasonicator.

Experimental methods and principles

Foam base fluid configuration. Different weight percent SDS and 1 wt % NPs were first added in 500 mL of DI water with the pH value adjusted. Then, the aqueous phase was mixed evenly using a stirrer (EUROSTAR 20 Digital) at a rotation speed of 1200 RPM for 120 minutes. After so, the nanofluids were transferred to a probe sonicator (SFX 550) for 120 min ultrasonication with a pulse on/off time of 5 s/2 s at 50% amplitude in an iced water bath. wash surfactant-coated nanoparticle (SNPs) three times with deionized water and ethanol alternately to obtain the final product.

Foam Generation and Stability Evaluation. The generation and stability of the foams were evaluated using a high-temperature, high-pressure (HTHP) foam generator, schematically illustrated in Figure 1. The apparatus, based on the Waring Blender method, characterizes foam properties (initial volume, half-life) under simulated reservoir conditions.

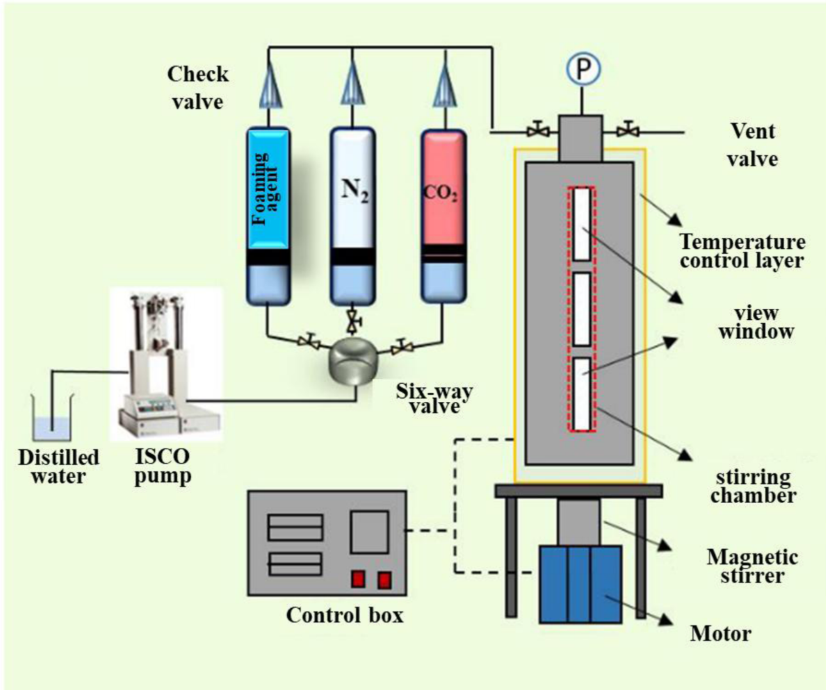

Figure 1—Foaming blender with high pressure and high temperature.

For each test, 50 mL of the foam base solution was injected into the HTHP blender cell. The cell was then pressurized to the target pressure with the specified gas phase. The gas and liquid phases were subsequently agitated by a magnetic-driven internal rotor operating at a constant speed of 1000 rpm for 6 minutes. After agitation, the initial foam volume (V) was recorded. The foam half-life ($t_1/_2$)—the time for the foam to release half its initial liquid—was measured as the primary stability metric.

Interfacial Rheology Measurement. The interfacial rheological properties—namely, the interfacial tension (IFT) and the interfacial dilatational viscoelastic modulus (E)—of the foam base solution against the gas phase were characterized using a Teclis Tracker-H interfacial rheometer based on the pendant drop method. The procedure involved first pressurizing the HTHP (High-Temperature, High-Pressure) cell with the gas phase (e.g., CO_2) to the desired temperature and pressure. A pendant drop of the foam base solution was then formed by injecting it into the cell. The profile of the gas-liquid interface was captured by a CCD camera. The system was allowed to equilibrate for 40 minutes to ensure a stable interface, after which the equilibrium IFT was measured. Subsequently, the interface was subjected to a sinusoidal oscillation of its area, with an oscillation frequency of 0.1 Hz and a volume amplitude of 10%. During this dynamic process, the transient IFT (γ) and the corresponding interfacial area (A) were continuously recorded. The interfacial dilatational viscoelastic modulus (E) was then calculated using the following equation:

$$E = \frac{\mathrm{d}\gamma}{\frac{\mathrm{d}A}{A}} = \frac{\mathrm{d}\gamma}{\mathrm{d}\ln A} \tag{1}$$

where E is the interfacial dilatational viscoelastic modulus (mN/m), γ is the interfacial tension (mN/m), and A is the interfacial area (m^2).

Foam Viscosity and Friction Factor Measurement. The effective viscosity and flow friction characteristics of the foam were evaluated using a custom-built high-temperature, high-pressure (HTHP) flow loop apparatus, which is schematically illustrated in Figure 2. This integrated system comprises a foam generator, a viscosity measurement section, and a friction testing loop. In the generation system, CO_2 and/ or N_2 are first compressed by a gas booster pump and then injected into a foam generator, where they are thoroughly mixed with the foam base solution. The gas flow rate is precisely regulated by a mass flow controller, while the liquid flow rate is delivered by an ISCO pump. This configuration enables accurate control over the foam quality and the gas phase composition by adjusting the respective flow rates. During each experimental run, the system was first allowed to achieve complete thermal and pressure stability within the circulation loop. The circulation pump and the data acquisition system were then activated. The fluid velocity within the pipe was controlled by the circulation pump. Once the readings from the flow meter and the differential pressure transducer stabilized, the steady-state flow velocity and the corresponding pressure drop were recorded. The relationship between flow velocity and pressure drop was subsequently used to calculate the foam's effective viscosity and friction factor, based on the principles outlined in Equations (2) and (3).

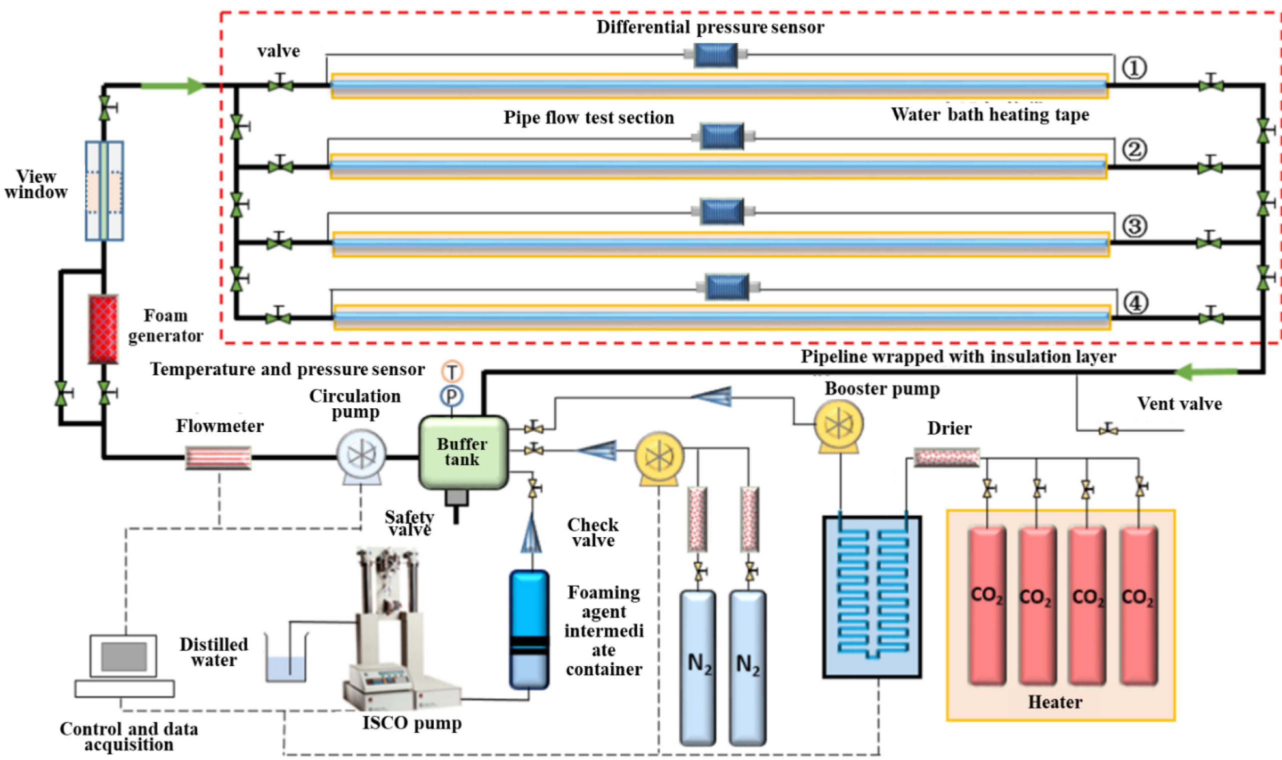

Figure 2—Schematic of large-scale tubular rheometer system.

$$\eta = \frac{\frac{\Delta p_f D}{4L}}{\frac{8u_{av}}{D}} \tag{2}$$

$$\lambda = \frac{2\Delta p_f D}{L\rho u_{av}^2} \tag{3}$$

where: η is the effective viscosity of the foam (mPa·s); Δp_f is the frictional pressure drop of the foam flowing through the straight pipe (MPa); D is the inner diameter of the pipe (m); L is the length of the pipe (m); u_{av} is the average velocity of the foam within the pipe (m/s); λ is the friction factor of the foam (dimensionless); ρ is the density of the foam mixture.

Results and Discussion

Interfacial Rheology

The macroscopic performance of foam fracturing fluids is governed by interfacial rheology, a critical parameter that dictates the rates of liquid drainage and gas diffusion (Ostwald ripening). This section investigates how N_2/CO_2 ratios influence interfacial properties to elucidate the synergistic stabilization mechanism. All experiments were conducted at 100°C, 30 MPa, and 80% foam quality.

Effect of N_2/CO_2 Mixture Ratio on Interfacial Tension. The gas/liquid interfacial tension (IFT), an indicator of surfactant adsorption efficiency, can be altered by the molecular packing of different gas components at the interface. Figure 3 illustrates the relationship between the equilibrium IFT and the proportion of N_2 in the gas phase at constant temperature and pressure. The equilibrium IFT increased slightly from 30.5 mN/m (pure CO_2) to 34.0 mN/m (pure N_2) as N_2 content increased (Figure 3). This limited impact occurs because IFT is predominantly determined by the surfactant's adsorption density and configuration, which were held constant. In this system, the type and concentration of the surfactant were held constant, preserving its dominant role in controlling IFT. The minor increase in IFT with higher N_2 content may be attributed to subtle differences in the intermolecular forces between the N_2/CO_2 molecules and the surfactant molecules at the interface, or to a slight disruption in the optimal packing efficiency of the surfactant.

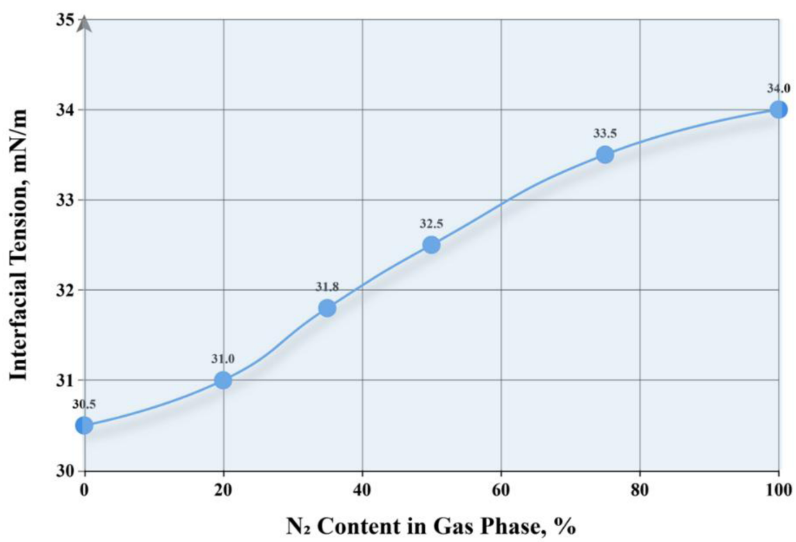

Figure 3—Effect of N_2/CO_2 mixture ratio on interfacial tension.

Effect of N_2/CO_2 Mixture Ratio on Interfacial Dilatational Viscoelasticity. The interfacial dilatational viscoelastic modulus (E) is a physical quantity that measures the ability of an interface to resist deformation and restore its original area when subjected to expansion or compression. A high dilatational viscoelastic modulus (E) indicates a strong self-repair capability (Marangoni effect), which suppresses liquid drainage and gas diffusion, leading to a highly stable foam. In contrast to IFT, the interfacial dilatational viscoelastic modulus (E) was highly sensitive to the N_2/CO_2 ratio, increasing dramatically with higher N_2 proportions (Figure 4).

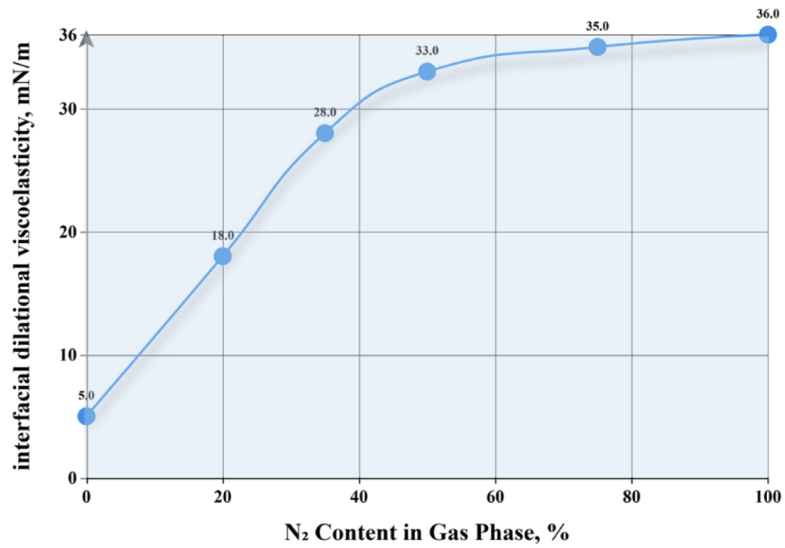

Figure 4—Effect of N_2/CO_2 mixture ratio on interfacial dilatational viscoelasticity.

In the pure CO_2 system (0% N_2), the interfacial dilatational viscoelastic modulus was at its lowest value, a mere 5.0 mN/m. This explains the inherent instability of pure CO_2 foams. When a lamella containing pure CO_2 is perturbed and stretched, the interfacial area increases, leading to a localized decrease in surfactant concentration and a momentary rise in interfacial tension. However, owing to the significantly higher solubility of CO_2 in water compared to N_2, dissolved CO_2 molecules from the bulk liquid phase rapidly diffuse to the newly formed interface, almost instantaneously compensating for the local pressure drop. This rapid mass transfer severely dampens the magnitude and persistence of the interfacial tension gradient, preventing the effective establishment of the Marangoni effect.

In the N_2/CO_2 mixed-gas systems, the introduction of N_2 fundamentally alters the mechanical response of the interface. N_2, as an inert gas with extremely low solubility in water, effectively acts as an "elastic backbone" or "mechanical anchor" at the interface. When a lamella of the mixed-gas foam is stretched, the local mole fraction and partial pressure of N_2 at the interface decrease instantaneously. Crucially, unlike CO_2, there is a negligible reservoir of dissolved N_2 in the bulk liquid available to rapidly replenish the interface. Consequently, a strong and persistent interfacial tension gradient, predominantly driven by the drop in N_2 partial pressure, is established and maintained. This powerful Marangoni stress acts akin to a "spring", effectively pulling liquid back into the thinning area to "heal" the lamella. While CO_2 still attempts to "relax" the interface via its dissolution-diffusion mechanism, its effect is largely suppressed by the robust elastic framework established by N_2. As a result, with an increasing N_2 proportion, the elastic response of the interface becomes progressively dominated by N_2, causing the dilatational viscoelastic modulus (E) to climb sharply. At an N_2 content of 50%, the value of E surged to 33.0 mN/m, representing a nearly six-fold increase compared to the pure CO_2 system.

Beyond an N_2 content of 35-50%, the rate of increase in E begins to decelerate, eventually entering a high-modulus plateau region. For instance, increasing the N_2 content from 50% to 100% only raises E from 33.0 mN/m to 36.0 mN/m, a markedly smaller increment. This indicates that once N_2 reaches a certain proportion in the gas phase, its contribution to interfacial elasticity becomes dominant, and the interface has already formed a sufficiently robust mechanical network to effectively resist external perturbations. Further increases in N_2 content yield diminishing marginal returns, as the interfacial modulus approaches the maximum achievable value for this particular surfactant system.

In summary, N_2 synergistically stabilizes the foam not by altering static IFT, but by leveraging its low solubility to enhance the interface's dynamic dilatational viscoelasticity. This transforms the "flaccid"

CO_2 interface into a resilient one with a strong self-healing capability (Marangoni effect), which is the microscopic origin of the improved macroscopic stability.

Analysis of Foam Microstructure. To further elucidate the mechanism by which the N_2/CO_2 ratio impacts foam stability at a microscopic level, the foam morphology generated from different gas systems was examined using microscopy. Figure 5 presents the microstructures of foams formed under constant temperature and pressure after the same initial agitation period (5 minutes). These images provide direct visual evidence for the macroscopic stability differences observed in the previous sections.

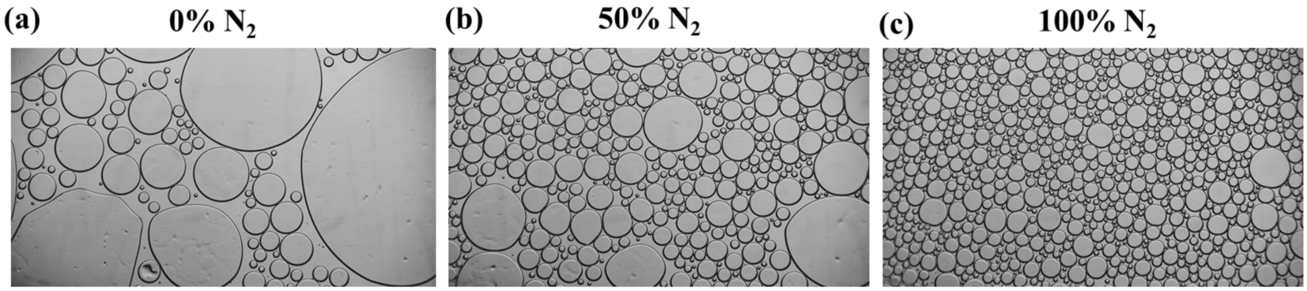

Figure 5—Microscopic morphology of foam lamellae with different N_2/ CO_2 ratios: (a) Pure CO_2 foam; (b) Foam with 50% N_2; (c) Pure N_2 foam.

The pure CO_2 foam (Figure 5(a)) shows a rapidly destabilizing microstructure, with a highly non-uniform, polydisperse bubble size distribution characteristic of vigorous Ostwald Ripening. This is a direct manifestation of vigorous Ostwald Ripening. Due to the high solubility and diffusion coefficient of CO_2 in the aqueous phase, gas molecules, driven by the Laplace pressure difference, rapidly diffuse from the high-pressure small bubbles to the low-pressure large bubbles. This leads to the rapid coarsening and subsequent collapse of the foam structure in a very short time. Upon introducing 50% N_2 into the gas phase (Figure 5(b)), the foam's microstructure undergoes a fundamental improvement. In stark contrast to the coarse, loose structure of the pure CO_2 foam, the mixed-gas foam is composed of bubbles that are significantly finer and more uniform in size. The bubbles are more tightly packed, showing a clear transition from spherical to polyhedral shapes, which forms a more stable network. The presence of well-defined Plateau borders indicates that the rate of liquid drainage from the lamellae has been effectively suppressed. This dramatic morphological transformation provides compelling evidence that the introduction of N_2 greatly mitigates the rate of Ostwald Ripening. The core mechanism is the "Trapped Gas Effect": as CO_2 attempts to diffuse out of a small bubble, the nearly insoluble N_2 molecules are left behind or "trapped", causing their partial pressure inside the bubble to rise. This, in turn, creates an osmotic pressure gradient that counteracts the outward diffusion of CO_2, effectively "locking" the foam structure in place. For comparison, the pure N_2 foam (Figure 5(c)) displays the most ideal and stable morphology. Its bubbles are the finest and most uniform, forming a well-defined, textbook-like polyhedral framework. This exceptional stability is attributed to the extremely low solubility of N_2, which fundamentally eliminates Ostwald Ripening, a primary destabilization pathway. A comparison of Figure 5(b) and 5(c) reveals that the uniformity and packing density of the 50% N_2 mixed-foam microstructure already closely approach those of the pure N_2 foam.

In conclusion, this visual progression—from the "highly polydisperse, coarse, and loose" structure of the pure CO_2 foam, to the "denser and more uniform" structure of the mixed-gas foam, and finally to the "highly uniform and ordered" structure of the pure N_2 foam—provides powerful microscopic evidence for the core mechanism by which N_2 synergistically enhances foam stability through the suppression of gas diffusion.

Foam Generation and Stability

As foam fluids are thermodynamically unstable systems, they cannot maintain constant properties over long periods. Therefore, for their application as fracturing fluids, it is crucial that they can be generated rapidly and subsequently maintain excellent stability during operations. This section compares the generation and stability characteristics of foam systems with varying N_2/CO_2 mixture ratios by investigating their foam volume and drainage half-life at different temperatures and pressures.

As shown in Figure 6, the foam volume exhibits a continuous and gradual increasing trend as the proportion of N_2 in the gas phase increases. The foam volume for the pure CO_2 system (0% N_2) was 400 mL, whereas for the pure N_2 system (100% N_2), it reached 525 mL, an increase of approximately 30%. The fundamental reason for this phenomenon lies in the direct influence of gas solubility on the foaming process. CO_2, being a gas with relatively high solubility in the aqueous phase, partially dissolves into the liquid during foam generation by agitation until saturation is reached, rather than being entirely utilized to form new gas/liquid interfaces. To some extent, this reduces the efficiency of gas conversion into the foam's gaseous phase for a given gas-liquid volume ratio, resulting in a smaller initial foam volume. Conversely, N_2 is an inert gas with extremely low solubility in water. When N_2 is injected into the foaming agent solution, nearly all gas molecules participate directly in the formation of bubbles. Consequently, for the same energy input, N_2 can be more efficiently dispersed into a large number of fine bubbles, yielding a greater foam volume. When N_2 and CO_2 are mixed, the overall solubility of the gas phase decreases as the N_2 fraction increases, allowing more gas to be "trapped" within the foam structure, which manifests macroscopically as a steady increase in foam volume. In contrast, foam stability (half-life) showed a non-linear and significant increase with the N_2/CO_2 ratio, revealing a strong synergistic stabilization effect between the two gases (Figure 6). The pure CO_2 foam (0% N_2) exhibits an extremely short drainage half-life and the poorest stability. This is consistent with the industry consensus and is rooted in two mechanisms. First, as discussed in the Interfacial Rheology section, the pure CO_2 interface possesses a very low dilatational viscoelasticity due to rapid gas mass transfer, leaving the lamellae with little self-repair capability and leading to fast liquid drainage. Second, and more critically, the high solubility and diffusion coefficient of CO_2 lead to a vigorous Ostwald Ripening process. Driven by Laplace pressure, CO_2 in the high-curvature small bubbles dissolves into the liquid lamellae and rapidly diffuses to the low-curvature large bubbles, causing the small bubbles to disappear and the large ones to grow. This results in rapid coarsening and collapse of the foam structure. The stability of N_2/CO_2 mixed-gas foams, however, is revolutionarily improved. As seen in Figure 6, when only 35% N_2 is incorporated into the gas phase, the foam's half-life increases by an order of magnitude, soaring from 20 min for pure CO_2 to 500 min—a nearly 24-fold improvement. The core mechanism behind this synergy is that the low-solubility N_2 molecules act as a gas diffusion barrier within the mixed-gas foam. As CO_2 attempts to diffuse out of a small bubble due to Laplace pressure, the nearly insoluble N_2 molecules are "trapped" inside. This causes the partial pressure of N_2 within the small bubble to rise sharply, creating an osmotic pressure gradient that opposes the Laplace pressure and thus powerfully suppresses the out-flux of CO_2. This phenomenon, known as the "Trapped Gas Effect", fundamentally slows the rate of Ostwald Ripening. Simultaneously, the introduction of N_2 significantly enhances the interfacial dilatational viscoelasticity, imparting greater resilience to the lamellae and further slowing liquid drainage caused by gravity and capillary forces. As the N_2 proportion is further increased, the drainage half-life continues to grow, gradually approaching that of the most stable pure N_2 foam. This indicates that once N_2 becomes the dominant component in the gas phase, the foam decay mechanism is entirely controlled by the extremely slow diffusion rate of N_2.

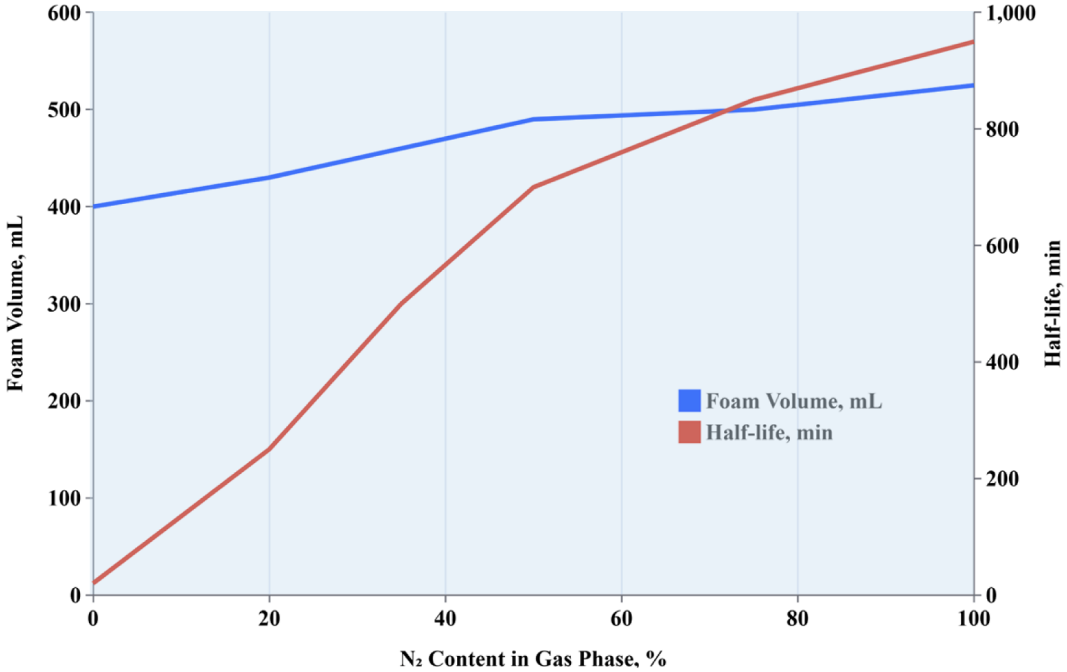

Figure 6—Foam volume and drainage half-life of foams with different N_2/CO_2 ratios.

In summary, the macroscopic properties of N_2/CO_2 mixed-gas foams exhibit a unique synergistic advantage. While foam volume increases moderately with N_2 content, the foam stability undergoes a non-linear and dramatic improvement upon the introduction of N_2. Particularly in the range of 35-50% N_2, the system not only achieves outstanding stability far exceeding that of pure CO_2 foam and approaching the level of pure N_2 foam, but it also retains the potential benefits of CO_2 (such as reducing the viscosity of reservoir fluids and improving interactions with reservoir rock). This provides a clear optimization pathway for the design of high-performance foam fracturing fluid systems.

Foam Rheology

The rheological properties of foam, particularly its effective viscosity, are critical parameters that determine its ability to effectively carry and transport proppant and to efficiently create fractures in the reservoir. High-viscosity foams provide superior proppant suspension capability and enable the formation of wider fractures. This section aims to systematically investigate the influence of the N_2/CO_2 mixture ratio on the rheological characteristics of the foam through laboratory rheological experiments.

Effect of N_2/CO_2 Mixture Ratio on Foam Effective Viscosity. Figure 7 displays the effective viscosity as a function of shear rate for foam systems with different N_2/CO_2 mixture ratios, under constant foam quality (80%) and thermobaric conditions. All systems exhibit typical shear-thinning behavior, meaning the effective viscosity decreases as the shear rate increases. This is a hallmark characteristic of foams as complex non-Newtonian fluids. At low shear rates, the internal bubble network of the foam remains intact, and the interactions between bubbles are strong, resulting in high viscosity. As the shear rate increases, the bubbles deform, elongate, and align in the direction of flow, causing the internal structure to be "disrupted", which reduces flow resistance and consequently lowers the viscosity.

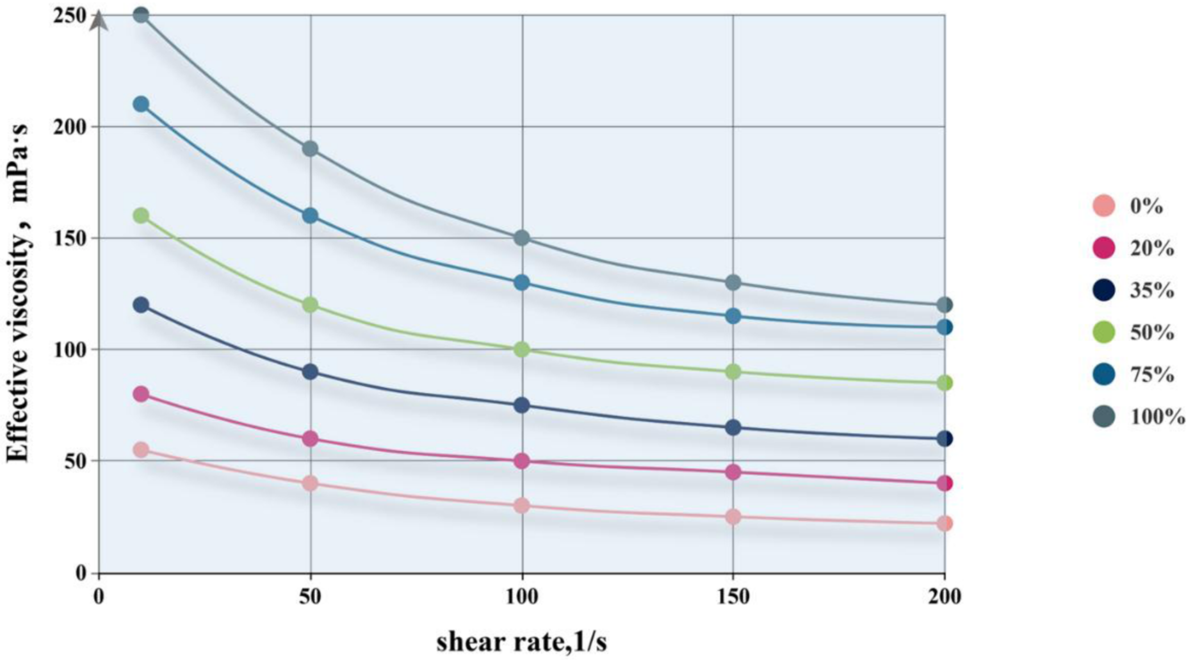

Figure 7—Effective viscosity of foams as a function of shear rate.

The N_2 proportion has a decisive, positive effect on the foam's effective viscosity. At any given shear rate, the effective viscosity follows the order: Pure N_2 foam > N_2/CO_2 mixed-gas foam > Pure CO_2 foam. At a shear rate of 10 s^{-1}, the viscosity of the pure CO_2 foam was only 55 mPa·s, whereas the viscosity of the 50% N_2 mixed-foam surged to 160 mPa·s (a nearly 3-fold increase), and the pure N_2 foam reached as high as 250 mPa·s (a 4.5-fold increase).

Pure CO_2 Foam (Source of Low Viscosity): As described in the previous section, pure CO_2 foam is extremely unstable due to vigorous Ostwald Ripening, resulting in a coarse microstructure with non-uniform bubble sizes. This fragile and loose internal structure is easily disrupted and reorganized under shear stress, failing to provide effective flow resistance, thus yielding the lowest macroscopic viscosity. N_2/CO_2 Mixed-Gas and Pure N_2 Foams (Source of High Viscosity): The introduction of N_2 significantly suppresses gas diffusion via the "Trapped Gas Effect", leading to the formation of a fine, uniform, and stable polyhedral foam structure. This ordered and resilient "bubble skeleton" possesses a greater resistance to deformation. When subjected to shear, the slip and rearrangement of bubbles must overcome a larger energy barrier, which manifests macroscopically as a significant increase in viscosity. The higher the N_2 content, the stronger this stabilization effect, the more robust the foam's "skeleton" becomes, and consequently, the effective viscosity systematically increases.

Having established the fundamental rheological behavior, we further investigated the synergistic effect between the gas composition and foam quality. Figure 8 presents the complete curves of effective viscosity as a function of foam quality for the six gas compositions. This plot is a central finding of this study, as it perfectly reveals the "operating window" of the N_2 stabilization effect. The entire process can be divided into three distinct regions: Region I: Low Foam Quality Zone (Foam Quality ≤ 50%). In this region, the bubbles exist as a dispersed phase, and a coherent foam structure has not yet fully formed. As shown, the viscosity values for all six curves are low and the differences between them are negligible. This provides strong evidence that at low gas content, the system's rheology is dominated by the continuous phase (the base fluid), and the influence of the gas composition is not yet apparent. Region II: Mid-to-High Foam Quality Zone (50% < Foam Quality ≤ 85%). This is the "golden zone" for foam performance and the core region where the differences between formulations become pronounced. As bubbles are compressed into a tightly packed polyhedral structure, gas diffusion becomes the dominant factor influencing stability. At this stage, the advantage of N_2 is dramatically amplified. As seen in the plot, the six curves rapidly "diverge" in this region. Systems with higher N_2 content exhibit a steeper slope in viscosity increase and reach a higher peak viscosity. The vertical separation between the curves is maximized in this zone, visually demonstrating the dominant role of the N_2 stabilization effect in this interval. Region III: Extremely High Foam Quality Zone (Foam Quality > 85%). When the foam quality becomes excessively high, lamella drainage and mechanical rupture become the dominant destabilization mechanisms. This is a physical process that is destructive to all systems. Consequently, we observe that the viscosity of all curves begins to drop sharply. A more critical observation is that the separation between the six curves rapidly diminishes, showing a trend of "convergence". This explains why, at very high foam qualities, the stability advantage conferred by N_2 is reduced—the system's "Achilles' heel" has shifted from chemical stability (gas diffusion) to the physical integrity of the lamellae.

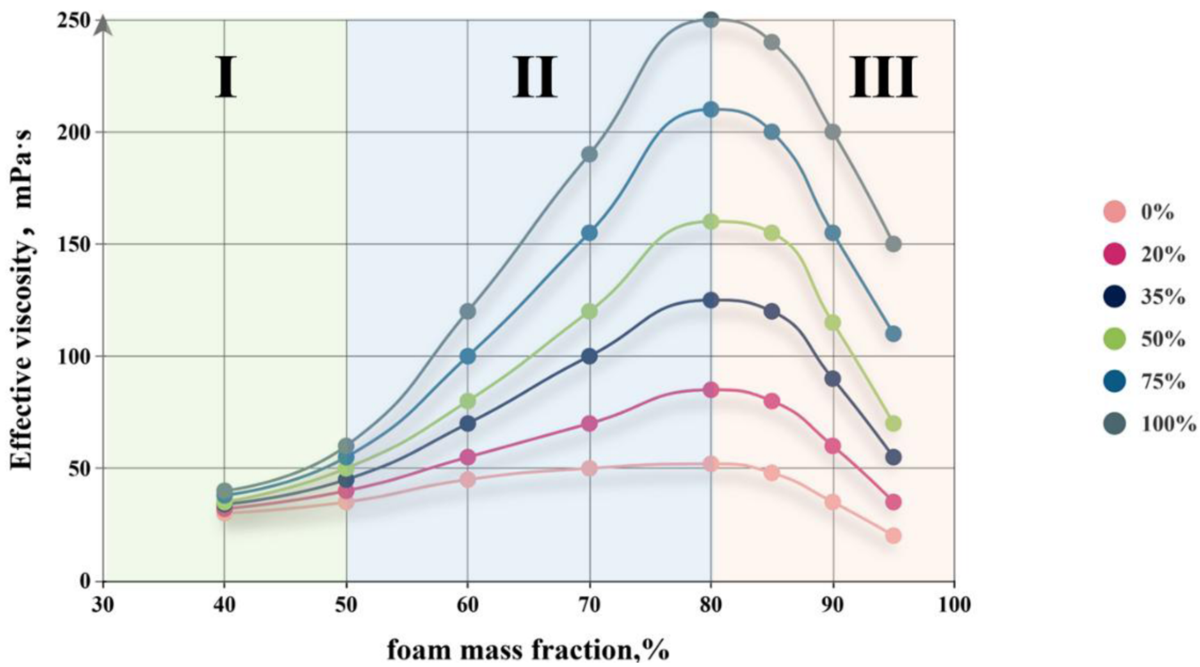

Figure 8—Effective viscosity of foams as a function of foam mass fraction.

Effect of N_2/CO_2 Mixture Ratio on Foam Friction Factor. To investigate the pipe flow characteristics of the foam, we first measured the friction factor as a function of average flow velocity at a fixed foam quality of 80%. Figure 9 illustrates the relationship between the friction factor and the average flow velocity for foams with different gas compositions at the same foam quality.

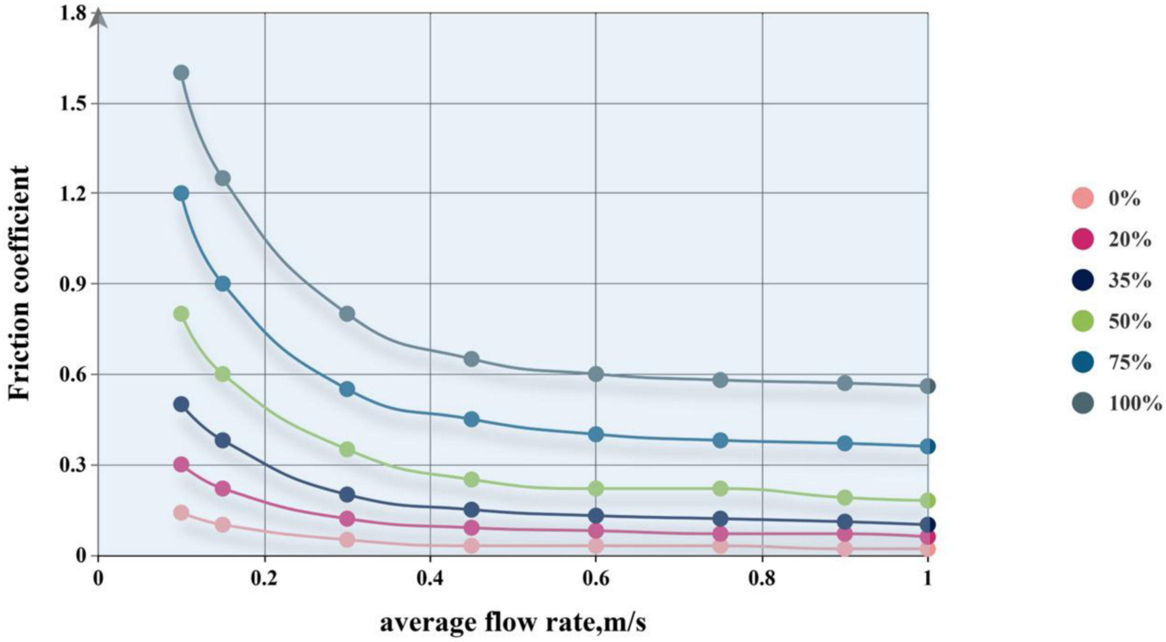

Figure 9—Friction factor versus average flow velocity for foams with different N_2/CO_2 ratios.

The results clearly indicate a strong positive correlation between the foam's friction factor and the N_2 content in the gas phase. For instance, at a flow velocity of 0.60 m/s, the friction factor for pure CO_2 foam was merely 0.03. This value increased to 0.22 (a more than 7-fold increase) when the N_2 content was raised to 50%, and soared to 0.60 (a 20-fold increase) for pure N_2 foam. The fundamental reason for this trend is the change in viscosity. According to the principles of fluid mechanics, the friction factor is proportional to the pressure drop within the pipe, which in turn is directly related to the fluid's viscosity. As established in Section *Effect of N_2/CO_2 Mixture Ratio on Foam Effective Viscosity*, increasing the N_2 content significantly enhances the foam's effective viscosity. When this high-viscosity fluid flows, the internal shear and the friction with the pipe wall are more intense, leading to greater energy loss and a larger pressure drop, which is directly manifested as a sharp increase in the friction factor. The increase in viscosity is the dominant factor driving the increase in friction.

To further elucidate the synergistic relationship between friction factor and foam quality, we measured the friction factor as a function of foam quality at a fixed average flow velocity (u_{av} = 0.60 m/s). The flow friction factor (λ) of a foam is directly related to its macroscopic viscosity. As depicted in Figure 10, the trend of the friction factor as a function of foam quality mirrors the effective viscosity trend (shown previously in Figure 8) with remarkable consistency. In Region I, all systems exhibited low friction factors with little difference between them. In Region II, the friction factor for systems with higher N_2 content increased dramatically, perfectly in sync with the divergence observed in viscosity. In Region III (foam quality > 80%), the friction factor decreased as the foam quality increased. This is because at excessively high foam qualities, the foam becomes unstable, potentially transitioning towards a mist flow regime. In this state, the foam's structural integrity is compromised, leading to a sharp decrease in viscosity and, consequently, a smaller frictional pressure drops. Although the foam's density also decreases with increasing foam quality (which would tend to lower friction), the change in frictional pressure is the dominant factor responsible for the overall decrease in the friction factor. This confirms that macroscopic viscosity, determined by foam quality and gas composition, is the core parameter governing frictional pressure drop.

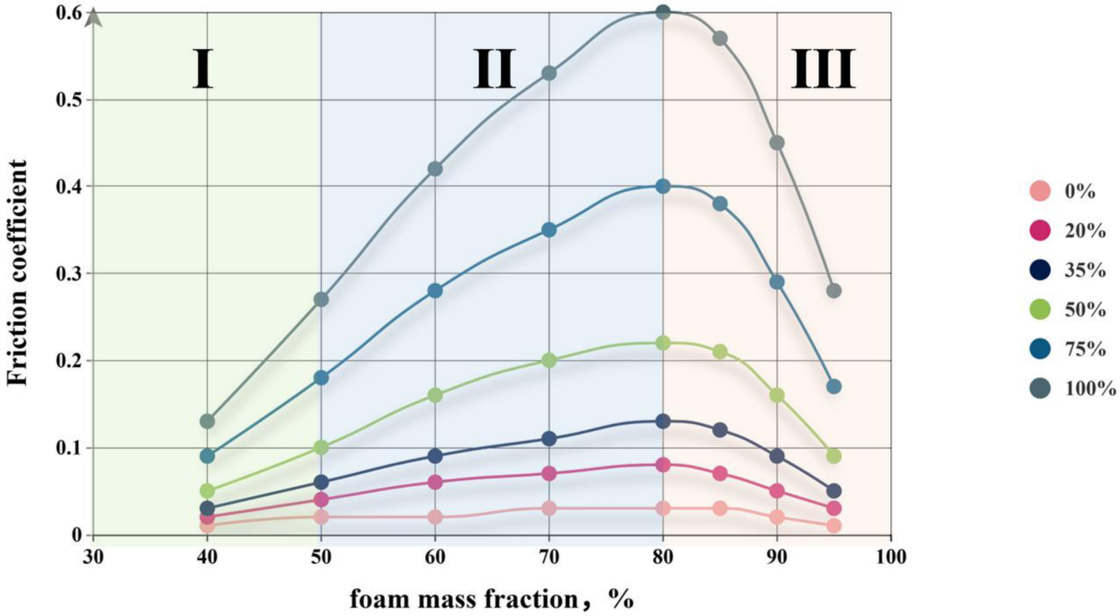

Figure 10—Friction factor versus foam quality for foams with different N_2/CO_2 ratios.

Comprehensive Discussion and Implications for Engineering Applications

A key engineering trade-off exists between performance (high viscosity for proppant transport) and cost (frictional pressure loss during pumping). This study quantifies this trade-off. From a performance standpoint: Achieving high viscosity necessitates the use of high-N_2 content formulations. According to the data in Figure 7 (rheology curves), at a shear rate of 10 s^{-1}, the viscosity of pure N_2 foam (250 mPa·s) is 4.5 times that of pure CO_2 foam (55 mPa·s), demonstrating a significant performance advantage. From a cost standpoint: Pursuing low friction implies using low-N_2 content formulations. According to the data in Figure 9 (friction factor curves), at an average velocity of 0.60 m/s, the friction factor of pure N_2 foam (0.60) is a full 20 times that of pure CO_2 foam (0.03). Such a massive increase in friction would pose a severe challenge to surface pumping systems. Thus, performance enhancement is inextricably linked to increased friction. For instance, a 50% N_2 foam offers a 3-fold viscosity increase but at the cost of a 7-fold higher friction factor, highlighting the necessity of optimization. The comprehensive gas composition–foam quality–rheology–friction database established in this study provides a direct, quantifiable basis for making such fine-tuned optimization decisions. Given the existence of this trade-off, identifying an "optimal balance point" becomes the central goal of the design process. Through a comprehensive analysis of the entire dataset, we can propose a clear design criterion: by holistically considering the performance gains against the friction cost, this study identifies the N_2 content range of 35% to 50% as the "optimal engineering window" for N_2/CO_2 mixed-gas foam fracturing fluids. The justification for this window is twofold: Performance Requirements are Met: Within this window, the foam's macroscopic properties are substantially improved. Taking the 35% N_2 formulation as an example, its drainage half-life already exceeds 500 minutes (nearly 24 times that of pure CO_2), providing sufficient stability for field operations. Its effective viscosity is also far superior to that of pure CO_2 foam, indicating good fracture creation and proppant transport capabilities. Costs are Relatively Controllable: Although the friction of foams in this window is higher than that of pure CO_2, the cost increase is within an acceptable range compared to pure N_2 foam. For example, while the friction factor of the 50% N_2 foam (0.22) is 7 times that of CO_2, it is only about one-third that of pure N_2 foam (0.60).

In conclusion, the key advantage of the N_2/CO_2 system is its "performance tunability". By controlling the N_2/CO_2 ratio, operational costs (friction) can be managed while meeting performance metrics (viscosity), enabling customized fluid design.

Conclusion

By employing a dual-scale experimental approach that integrates macroscopic and microscopic analyses under simulated deep reservoir conditions (100°C, 30 MPa), this study systematically evaluated the performance of N_2/CO_2 mixed-gas foam fracturing fluids and elucidated their synergistic enhancement mechanism. The outcomes of the study are summarized as follows:

1. Microscopic Mechanism: N_2's synergistic enhancement stems from its low solubility, which increases the interfacial dilatational viscoelastic modulus (nearly 6-fold with 50% N_2) and suppresses Ostwald Ripening via the "Trapped Gas Effect." This enhances the Marangoni effect and dramatically improves macroscopic foam stability.
2. Macroscopic Performance: Adding N_2 significantly improves performance. With 35% N_2, the foam half-life increased 24-fold (from 20 to >500 min). With 50% N_2, effective viscosity nearly tripled (to 160 mPa·s at 10 s^{-1}), sufficient for proppant transport.
3. Engineering Trade-off: A clear trade-off between performance and cost was defined. Enhanced performance leads to higher frictional pressure; for instance, the friction factor for 50% N_2 foam is over 7 times that of pure CO_2 foam. This relationship necessitates optimization, for which the study's comprehensive "gas composition–foam quality–rheology–friction" database provides a quantitative basis.
4. Optimal Design Guideline: An N_2 content of 35-50% is proposed as the "optimal engineering window." This range balances superior stability and viscosity with manageable friction costs, achieving an optimal trade-off between performance and engineering feasibility.

Research Outlook

Future work should validate these findings under more representative reservoir conditions. Key investigations should include assessing the foam's interaction with crude oil in saturated cores and determining the effect of proppant addition on the system's rheology and stability.

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